

Verbal vs. Visual: How Humans Perceive and Collaborate with AI Teammates Using Different Communication Modalities in Various Human-AI Team Compositions

RUI ZHANG, Clemson University, USA

WEN DUAN, Clemson University, USA

CHRISTOPHER FLATHMANN, Clemson University, USA

NATHAN MCNEESE, Clemson University, USA

BART KNIJNENBURG, Clemson University, USA

GUO FREEMAN, Clemson University, USA

As AI becomes more advanced in natural language processing, the research on AI's communication in teaming environments is getting more attention in CSCW/HCI. Even though AI's communication serves as an essential part of facilitating team coordination and shaping team outcomes, the impact of AI's communication modality on human-AI teamwork is still understudied. Using a mixed-design experiment and follow-up group interviews with 100 participants, we investigate the impact of AI's communication modality, one of AI's most essential communication characteristics, on team coordination and team outcomes in two different human-AI team compositions, human-human-AI teams, and human-AI-AI teams. Our findings highlight the trade-offs between AI's verbal communication and visual communication, which inspire two design recommendations on how to apply AI's verbal and visual communication to support human-AI coordination effectively. Our study generates an initial understanding of the role that AI's communication will have in ensuring team effectiveness in human-AI teams in the CSCW/HCI field.

CCS Concepts: • **Human-centered computing** → **Collaborative and social computing**; **Human computer interaction (HCI)**.

Additional Key Words and Phrases: AI's communication modalities; human-AI communication; team composition; human-AI teaming

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1 Introduction

Driven by advanced machine learning technologies, the design and integration of artificial intelligence (AI) as a collaborative teammate with humans have garnered considerable interest [63, 66, 127]. This pursuit aims to achieve unprecedented team outcomes by harnessing the potential of human-AI teams (HATs). Since communication is the primary method through which

Authors' Contact Information: Rui Zhang, rzhang2@clemson.edu, Clemson University, Clemson, South Carolina, USA; Wen Duan, Clemson University, Clemson, South Carolina, USA, wend@clemson.edu; Christopher Flathmann, Clemson University, Clemson, South Carolina, USA, cflathm@clemson.edu; Nathan McNeese, Clemson University, Clemson, South Carolina, USA, mneese@clemson.edu; Bart Knijnenburg, Clemson University, Clemson, South Carolina, USA, bartk@clemson.edu; Guo Freeman, Clemson University, Clemson, South Carolina, USA, guof@clemson.edu.



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team members coordinate and collaborate over shared tasks and develop a sense of team [100], understanding the *communication* needs and designing for supporting these needs is critical for designing effective HATs.

An integral aspect of team communication is modality. Myriads of past CSCW and CMC research have demonstrated the impact of communication modality on team processes and outcomes in *human-only teams* (e.g., [35, 62]). For instance, compared to leaner communication modalities (e.g., text-based verbal communication), some other richer communication modalities (e.g., voice-based, video-mediated communication) have been found to better foster team trust and cooperation [55], enhance coordination [39, 62] and team performance [33] for various types of tasks and teams [67]. However, different communication modalities involve different trade-offs. While text-based communication has disadvantages such as a difficulty in managing large-volume group conversations [115], it affords reviewability [35], which is vital for teams and organizations to refer back to for transparency and accountability purposes [53]. Yet, nonverbal communication may be susceptible to different interpretations causing ambiguity and miscommunication, and is often transient, leaving no trace for reference [79].

Currently, a vast majority of empirical research on HAT employs only verbal text communication [26, 27, 76] with only a small number of studies allowing communication in other modalities (e.g., behavior-based nonverbal communication) [48, 60]. For instance, previous work has explored how agents' verbal text-based communication and behavioral communication (i.e., communicating by taking actions) impacts shared mental models in human-AI teams [48]. Given the importance of nonverbal communication to teamwork [36, 39, 55], the scarcity of empirical HAT research that leverages nonverbal cues is unfortunate. Due to this dearth of empirical work, little is known about how different communication modalities impact team coordination processes and team outcomes in HATs, let alone their interaction with team context. This potential ignorance of alternative communication modalities hinders the effective design of AI communication capabilities [77]. Thus, more research is needed to understand how AI's communication modalities impact team coordination and then how AI's communication modalities should be applied to maximize the communication component in teamwork for desired team outcomes.

To address this gap, we explore how communication modalities (verbal vs. visual) differently impact human perceptions and team outcomes for HATs, and how the trade-offs of verbal and nonverbal communication play out in these teams through a mixed methods experiment with 100 participants. This study addresses the following research questions:

RQ1 How do AI teammate communication modalities impact team coordination processes and team outcomes across different team compositions in HATs?

RQ2 How do people perceive AI's verbal vs. visual communication in HATs similarly or differently?

Specifically, RQ1 explores the impact of AI's communication modalities (i.e., text-based verbal and visual communication) on team collaboration processes, including task-related performance and team outcomes across two HAT team compositions, human-human-AI (HHA) teams and human-AI-AI (HAA) teams. We focus on these team compositions because they have been found to impact humans' trust in AI teammates [104], the fostering of which requires communication. RQ2 focuses on understanding the similarities and differences in how humans interpret AI's verbal vs. visual communication using interview data.

This work makes three contributions to CSCW and HCI research. First, this study provides new insights into the trade-off of AI applying verbal vs. visual communication in supporting team coordination processes in HATs. This study uncovers that AI's visual communication decreases participants' mental workload, and benefits team performance in HAA teams, even though these benefits were not perceived as such. This work serves as a foundation for future CSCW and HCI

research on AI teammate communication to further understand how to design effective AI communication that facilitates human-AI coordination in HATs using both verbal and visual communication. Second, our study expands the current CSCW and HCI research on human-AI teaming [66, 128] by shedding light on how humans perceive AI using verbal vs. visual communication in team collaboration. This understanding is important in developing an AI teammate that meets humans' communication expectations and can coordinate with humans to accomplish high-complexity team tasks. Lastly, this study offers valuable insights into the design of communication capabilities for AI teammates that effectively harness the trade-offs between visual and verbal modalities.

2 Related Work

2.1 The Importance of AI's Communication Modalities

Communication is not only a critical consideration for humans but also for AI systems, as research has continuously sought to design machine systems that can communicate and collaborate with humans. While computational efforts have rapidly increased the ability, accuracy, and efficiency of AI algorithms [125], these performance boons are a somewhat moot point if these newfound abilities do not bolster human performance. As such, in addition to exploring the ability of AI to complete tasks, research has also had to explore the communicative capabilities of AI systems [16, 38]. In turn, the ability to communicate often stands as a key facilitator in enabling AI to collaborate and share information with humans [45, 121]. Recently, this benefit has become especially apparent with the recent introduction of highly-performative large language models, which have superb natural language capabilities, as they have rapidly shown the potential for AI systems in a variety of domains [10, 11, 58]. However, verbal conversation is not the only way AI systems can communicate. In reality, there are often two distinct modalities through which AI systems can communicate: verbal conversation and nonverbal displays. Critically, each modality presents its own benefits and unique design considerations to ensure optimal AI communication.

For verbal conversation, AI systems often leverage natural language communication capabilities to drive the conversation. While these capabilities have been researched for decades [21, 84], recent advancements have seen commercially available AI systems leverage language capabilities that can more accurately replicate human language [43]. However, good verbal communication from AI is not solely defined by its replication of human language. Rather, AI systems that communicate via verbal modalities need to be considerate of context [129], desired levels of anthropomorphism [94], etiquette [86, 93], and privacy [118]. As such, the actual form of AI conversation has to change based on these factors, which results in the design and implementation of verbal communication requiring a more local optimum than global optimum approach. However, when this local optimum is taken, and AI verbal communication is designed correctly, it can lead to demonstrable benefits to both the performance of human-AI collaborations and the perceptions humans have for said AI, such as trust [54, 96]. Critically, these positive perceptions can enable verbal communication to have actual behavioral outcomes for humans within these collaborations. For instance, conversational AI in both health and corporate business domains have significantly increased humans' engagement with health organizations and businesses [54].

Importantly, the prevalence of research on verbal communication by AI has not reduced the amount of research that explores the importance of nonverbal communication. In reality, research has continuously shown that nonverbal communication is just as important to AI as verbal communication, especially when embodied AI systems are of concern, such as robotics or digital agents [2, 17, 65]. For instance, eye contact and eye communication have been shown to be critical considerations when designing digital agents that interact with humans [7, 124], and these benefits have even been shown to translate to robotic systems as well, despite robotic systems not always

being anthropomorphic [65]. Similarly, gestures are a critical affordance of embodied agent systems as they can enable the formation and recollection of heuristics that enable humans to quickly interpret an agent's intent and act around the agent [14, 101]. These benefits have even been shown to exist in contexts where verbal communication is often prioritized, such as in socially assistive robotics [1]. Moreover, the benefits of nonverbal communication often transcend unidirectional communication from AI to humans, as understanding nonverbal communication can help create bidirectional communication streams between humans and AI where both entities can interpret and communicate intent through nonverbal means [57, 98].

In sum, there is no universally optimal communication modality for AI systems. Rather, designers and developers have to continuously monitor contextual and individual human needs when crafting the communicative design of AI systems [129]. In particular, while each individual modality requires unique design considerations, one must first consider which modality or combination of modalities is ideal for a particular AI system [96]. As such, there is a need within human-AI teams to understand the impacts and roles that these varying communication modalities have on individual human and team outcomes.

2.2 Human-AI Teams, and the Relevance of Communication and Composition

Human-AI teams represent a novel and continuously advancing team construction and application of AI technology. These teams leverage AI systems in an interdependent role with humans, where this shared interdependence is used to drive the efficient completion of shared goals [91]. When operating to their fullest extent, these teams have the ability to perform similarly to if not exceed human-human teams [6, 75], especially when the context of these teams allows humans and AI to intelligently complement each other [52]. However, the effectiveness of human-AI teams is not a direct result of maximizing the individual performance and contribution of AI teammates, as greater performance and contribution can negatively impact the performative potential of human teammates [5, 32]. In turn, AI teammates that are advanced in their skills in isolation do not always see said individual performance translate to team effectiveness.

This sentiment also pertains to the concept of communication in human-autonomy teams, as team communication requires a level of intelligent design; however, existing research on verbal human-AI team communication is still somewhat limited. Currently, research has shown that both the frequency of language [56] and the etiquette of language [31] directly impact the perception and performance of human-AI teams. Notably, both verbal and nonverbal communication in these teams presents a potential means of human-AI and human-human communication streams matching each other, potentially making coordination and collaboration easier [3]. For instance, AI teammates have been shown to benefit team performance when they visually communicate spatial information [20, 103], and AI teammates have even been theorized as being used as a conduit that persistently scans and visually communicates environment information to humans [29]. In turn, AI teammates have the potential to leverage both verbal and nonverbal communication, with the former likely benefiting coordination and collaboration and the latter likely benefiting the awareness humans form in tasks.

Thus, AI teammate communication research needs to consider the needs of specific teams to determine when nonverbal or verbal communication should be leveraged. In particular, the concept of team composition, which is generally impactful to human-AI teams [78], is likely to be an important factor of consideration. Specifically, researchers and designers will need to consider the individual impacts of different communication modalities in teams, and how these impacts align with communication goals. For instance, teams in digital environments often leverage verbal communication to positively impact the exchange of complex information [25] and nonverbal information to positively impact coordination [64]. While research has yet to explicitly explore

these different modalities in human-AI teams, human-robot teams have seen similar impacts as well, with verbal communication helping team planning [89] and nonverbal communication helping real-time coordination [13]. However, these impacts will need to be evaluated based on their compatibility with existing communication in human-AI teams. For instance, the benefits of verbal communication might be especially positive in human-AI teams where multiple humans are present and verbally communicating. Nevertheless, nonverbal communication is likely to be more beneficial in teams with increasing numbers of AI teammates, such as swarms, as the prevalence of too much verbal communication could be cognitively burdensome [99]. Yet, minimal work has been done to explicitly and empirically explore the impact of composition and communication on human-AI teams, and no work has been done to explore the interaction of both on team effectiveness.

In evaluating the effectiveness of both communication and composition in human-AI teams, it is also important to understand how the effectiveness of these teams is actually determined. Generally, human-AI teams require the evaluation of both task performance and perception, as high task performance does not always equate to positive perceptions of AI teammates [4]. In particular, perceptual data often centers around workload, perceived performance, and trust [111], which research is continuously showing to be fundamental to human-AI teams [74]. Trust and workload are both especially important to this work, as the design of nonverbal AI teammate communication has been shown to impact trust [96, 119], and the potential for communication to cognitively burden humans is not unlikely. Moreover, qualitative methods are often employed to evaluate and understand these teams, as the novelty of AI teammates means that research goals and tools may not always fully align with humans [8]. In turn, mixed-methods work is often seen as highly beneficial within the human-AI teaming domain [46, 113], as empirical impacts can be quantified and then inductively detailed.

In summary, while both verbal and nonverbal communication is likely to play roles in human-AI teams, research needs to empirically evaluate these roles as well as their interaction with other teaming considerations, such as composition, and a mixed-methodology approach is likely ideal for this exploration.

3 Method

3.1 Experimental Design

This study employs a 2x2 mixed experimental task design (see Table 1). The two manipulations are (1) the communication modalities of AI teammates, manipulated as AI using verbal and visual communication; and (2) the composition of triad teams, including two conditions - human-human-AI (HHA) team and human-AI-AI (HAA) team.

Table 1. Experimental Design.

Team Composition (Within-subjects)	AI's Verbal vs. Visual Communication	
	Verbal Communication + HHA	Visual Communication + HHA
	Verbal Communication + HAA	Visual Communication + HAA

AI's communication modalities were designed to be a between-subjects manipulation, whereas team composition was designed to be a within-subjects manipulation. A within-subjects design was selected for team composition since it has better statistical efficiency than a between-subjects design by removing subject variances from error terms to manipulation effects [41]. The order of the within-subjects manipulation was randomized and counterbalanced to avoid order effect [18]. A

repeated measure design was used for each within-subject condition, where participants completed two, five-minute tasks with each team composition (e.g., HHA or HAA), resulting in four total tasks and post-task surveys. Two participants participated simultaneously in each experiment. They worked in separate teams in the HAA condition, and the same teams in the HHA condition. After participants completed four rounds of tasks and all the survey measurements, they were interviewed together regarding their perceptions of AI teammates' communication and their coordination with the AI across four rounds of tasks. On average, group interviews lasted twenty minutes.

3.2 Task Platform

For this study, we used *Rocket League* as the experiment platform. At a high level, Rocket League presents a team-based sports game where players use vehicles to play a simplified soccer game (see Figure 1 for game interface). In general, prior research in human-AI teaming has begun to highlight the utility of Rocket League as a research platform, as multiple prior studies have used it for empirical research [32, 117]. These prior studies have highlighted how Rocket League leverages an abstracted teamwork task that builds on existing teamwork principles while also providing a staggering variety of AI teammates due to community-supported AI capabilities [117]. In turn, Rocket League provides an ideal platform for human-AI teaming research, as teams are required to constantly coordinate around each other to complete a shared goal [32].



Fig. 1. Rocket League Game Task Interface

Regarding communication, Rocket League provides an especially ideal platform for this empirical research study. First, Rocket League actually has a technical glitch that reduces technical performance when bi-directional voice chat is used, which has led to players to focus on facilitating coordination with minimal communication. In particular, teams can leverage unidirectional push communication, where pre-made messages can be sent to signal common actions, such as going for a ball or falling back to play defense. Simultaneously, teams within Rocket League can consistently form awareness for each other, and leverage this awareness to anticipate teammate actions, goals, and strategy. In turn, both verbal communication components could benefit human-AI teams within Rocket League to facilitate push communication, but enhancing visual elements to more explicitly communicate teammate actions and strategy could do the same. Thus, while Rocket League does provide a somewhat limited scope in abstracting this study to a sports task, its relevance to human-AI teaming and communication makes it ideal for this foundational study.

Regarding the actual gameplay of Rocket League, three general parts are involved, each benefiting from intelligent communication and team composition decisions. Part one, kickoff, has teammates placed randomly on the soccer field facing a singular soccer ball, and teammates have to determine who should go for the soccer ball. During this part, communication is critical as it can be used to signal and decide who should go for the ball. After kickoff is the second part of the game, where teams control the ball toward their opposing teams' goal. Teams need to communicate to facilitate passes to each other and organize each others' positions, and this task can become demonstrably more complex based on the team composition, as coordinating multiple entities can create a cognitive burden. Lastly, teams are tasked with hitting the ball at their opposing team's goal. This final part is heavily predicated on communication as teammates need to coordinate in the case of a missed or blocked goal, which can be somewhat hectic and potentially confusing. In sum, team members' communication and team composition stands as critical considerations throughout the Rocket League task, and, as such, AI teammates' communication would likely benefit the task progress; however, these benefits may vary based on the composition of HATs.

3.3 AI Agents/Teammates and Their Communication Operationalization in Rocket League

The Rocket League community has built a platform called RLBot to create and share Rocket League bots for users to collaborate with offline. This RLBot platform (see Figure 2) includes bots that are equipped with various play strategies and features. In this study, we selected an AI teammate designed for team play and modified it to meet our research needs. Additionally, human teammates used only verbal communication (e.g., the text-based chat channel) to communicate with other teammates (i.e., both humans and AI teammates), regardless of AI's communication modality.

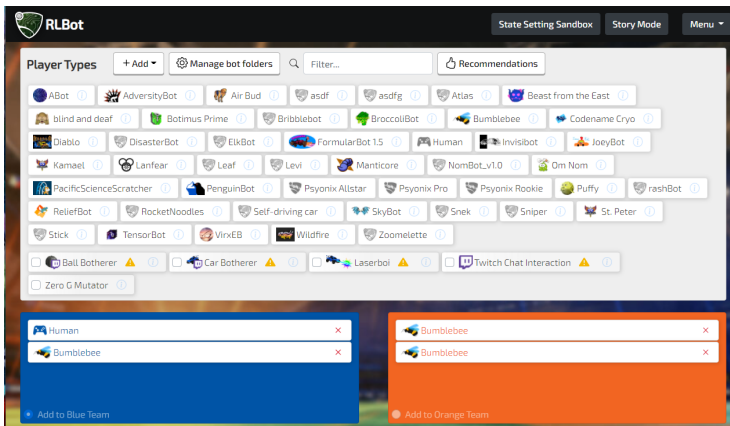


Fig. 2. RLBot Interface

3.3.1 AI's Verbal Communication. AI's verbal communication was operationalized using a text-based communication channel that allowed AI to share information with the team. AI teammates sent text messages to inform human teammates of their next steps (see Table 2). This communication script was developed based on AI's actions as triggers for specific dialogue options. Messages and corresponding triggered events (i.e., maneuvers) listed in the script are pre-defined and coded in AI's algorithms. One example of such triggered events is that the AI agent would send a message "Taking the shot!" in the group channel when their next algorithm-generated action is to shoot

the ball. The text-based communication channel is located in the middle of the screen (see Figure 3). This communication channel is also used for participants to communicate with other team members.

Table 2. AI’s Messages in the Verbal Communication Condition

Maneuvers	Messages
Strike	Hitting the ball!
Dribbling (carry and flick)	Dribbling the ball!
Close shot	Taking the shot!
Drive backwards to goal	Defending the goal!
Pickup a boostpad	Going for boost!
Kickoff	Going for kickoff!



Fig. 3. AI’s Text-based Verbal Communication

It is important to note that the AI was not equipped with the ability to parse humans’ text messages in order to verbally or behaviorally respond accordingly. In other words, verbal communication between human and AI teammates was not made bi-directional. While bidirectional communication has been found to be essential for human-AI teaming [109], given the current state-of-the-art AI technology, HAT research rarely allows for genuine bi-directional communication unless by employing the Wizard of Oz technique [24]. In some research, the Wizard even mimicked the communication limitation of AI to be unable to comprehend natural language to pass off as a real AI agent [106]. It is not uncommon for HAT research and practical HAT applications to have the AI perform unidirectional communication to inform humans about its decision-making process and outcomes [61, 68, 95, 106].

Our design decision for the one-way communication is a careful balance between genuine technical implementation, realism, and experimental control. First, this design aligns the directionality of verbal communication with that of visual communication, allowing us to solely focus on communication modality, canceling out the role played by communication directionality. Second, by restricting the AI teammate from responding to human requests, we maintain a higher level of experimental control. Allowing the AI to act on participants’ requests could introduce confounds stemming from variations in the presence, absence, and quality of these requests. Third, instead of

employing the Wizard of Oz technique, we opted to use computerized bots to maximize realism. This decision was made considering the complexities of synchronizing the AI’s task execution with natural language processing, striving to create a realistic scenario within practical limitations.

3.3.2 AI’s Visual Communication. AI’s visual communication utilized the trajectory of their predicted next steps to share their game status. This trajectory in the 3D space in the game is shown as a green straight line (see Figure 4). The next steps are calculated based on the maneuvers and events the AI is actively performing. Importantly, to make sure these two communication modalities (verbal vs. visual) present the same information, we designed the trajectory to be plotted only when AI teammates in the verbal condition would send a corresponding message. In other words, the triggered events for plotting a trajectory or sending a text message are the same, as listed in Table 2. A comparison between AI verbal vs. visual communication is presented in Table 3 to better present the difference between the two communication modalities. Moreover, each AI teammate in RLBot has a specific play strategy coded in the algorithm. By modifying a singular AI teammate, all the AI teammates (i.e., in both HHA and HAA teams) used within this study were able to be based on the same algorithmic foundation, which ensures that no team composition or communication modality varied in the behavioral actions of the AI teammates.



Fig. 4. AI’s Visual Communication

Table 3. Differences Between AI’s Verbal and Visual Communication

	Verbal	Visual
Triggered events (maneuvers)	Same (see Table 2)	
Communication based on triggered events	A text message (see Figure 3)	A straight line (see Figure 4)

3.4 Recruitment and Participants

This study used a department subject pool in a local university to recruit participants. Before the recruitment and data collection, the study protocol (including the recruitment approach, data collection methods and procedure, data management plan, and data analysis methods), survey measurements, interview protocol, and consent forms were submitted to the university’s Institutional Review Board (IRB) for review. Modifications were made based on IRB’s comments and suggestions.

After the modified study application was approved by the university IRB, the study was posted in the department subject pool for recruitment. Participants received course credits after they participated in the study and completed the tasks. The only recruiting criteria was participants had to be 18 years old or older. Based on several power analyses conducted for a 2x2 mixed-effect design, a sample size of 100 participants (e.g., 50 teams) is needed to meet the power of 0.85. Following the power analysis, 100 participants' valid data was collected, leading to 100 participants for each team composition (i.e., 50 HHA teams and 100 HHA teams), and 50 participants for each communication condition (i.e., verbal vs. visual).

72 participants (72%) identify as women, and 28 participants (28%) identify as men. It is important to note that gender representation in games is often imbalanced [37]. Previous work using gaming as a context usually has a substantially higher percentage of man participants than woman participants [83, 122]. Compared to these studies, this study provides a better understanding of how women perceive AI teammates and collaborate with them. Additionally, while this study's participants only reported men and women, non-binary was indeed provided as an option in the demographic question. Regarding gaming experiences, 68 participants (68%) had never played Rocket League before this study, and 32 participants (32%) had played Rocket League before. Other demographic information of participants is summarized and presented in Table 4.

Table 4. Demographic Information of Participants

Gender	Age	Ethnicity
		Asian- 4
Female- 72	range from 18 to 24 (mean = 18.85)	Black or African American- 8
Male- 28		Non-Hispanic White- 81
		Hispanic and Latino- 5
		Other- 2

3.5 Procedure

Prior to the study, five pilot studies were conducted to explore potential issues in the survey design, experiment tasks, and interview questions; test the average length of the experiment, and ensure that participants could fully understand the experiment process.

The actual study started with a consent form and a pre-survey that collected participants' demographic information. Participants then completed two rounds of training before the actual team tasks. The first training session was an individual training tutorial provided by Rocket League to get familiar with game controllers and operations. In the second training session, participants were asked to team up with an AI teammate that utilized either verbal or visual communication to complete a five-minute task. Participants were told that the AI teammate would communicate with them using a list of messages (for the verbal communication condition) or using a trajectory that indicated their next steps (for the visual communication condition). No specific roles were designated for any team member since this allows participants to develop their own form of team coordination with other teammates. This training also included teaching participants how to communicate with their teams using the text-based chat channel and were asked to use the messages provided in Table 5 for both AI's visual and verbal communication condition.

Once participants completed the training sessions, they were asked whether they had questions and whether they were ready to start the actual tasks. There were four rounds of five-minute tasks in total, including two rounds where participants worked together with one AI teammate (HHA team composition), and two rounds where participants worked in separate teams with two AI teammates

Participants' Messages List
Going for boost!
Defending the goal!
Taking the shot!
Hitting the ball!
Dribbling the ball!
Going for kickoff!
Distracting the opponent!

Table 5. Participants' Communication Script

each (HAA team composition). The order of the two team compositions was randomized to reduce the order effect [102]. Participants completed a post-survey to measure their perceptions after each round of tasks. Specific measurements are discussed in the next subsection. Once participants completed all the experiment tasks, they were interviewed together as a two-person group about their experiences with the AI teammate throughout the tasks.

3.6 Measurements

This study's measurements include three main parts. First, demographic information (e.g., gender and previous experience with Rocket League) was collected in the pre-survey. Second, post-measurements were used to collect participants' perceptions of the human/AI teammates and overall collaboration after each round of the task. These measurements are described below. Third, team performance was objectively measured using Rocket League's scoring system, which is described below. It should be noted that some measurements were taken at a teammate level, such as trust in the teammate, including trust in *both* teammates, whereas some were measured at a team level, such as team viability.

3.6.1 Demographic Measurements. Participants' demographic information (e.g., participants' age, gender, ethnicity, and education information) and previous Rocket League experience were first collected in the pre-survey. Their existing attitude toward AI was then measured using a 14-item scale adapted from a prior-developed scale- the Negative Attitude toward Robots Scale (NARS) [90], which has been verified and applied in multiple studies [112, 116]. This scale uses a five-point Likert scale, ranging from "Strongly Disagree" to "Strongly Agree". Example questions include "*I would feel uneasy if Artificial Intelligence (AI) really had emotions.*" and "*I would feel uneasy if I was given a job where I had to use AI.*" It should be noted that a higher NARS score indicates more negative attitudes (with 3 being neutral). In addition, participants' previous experience with Rocket League was categorized as experienced (i.e., who have played it before), and who had never played Rocket League before this study.

3.6.2 Teammate Measurements.

Perceived Teammate's Communication Quality. We developed a three-item measurement for this study to measure participants' perceived communication quality of their teammates. This scale uses a five-point Likert scale, ranging from "Strongly Disagree" to "Strongly Agree". One example question is "*I understood my AI/human teammate's communication.*" This measurement was taken twice after each round of the task, once for each teammate. Cronbach's alpha was applied to ensure the internal consistency and reliability of this measurement. As shown in Table 6, Cronbach's alpha of the perceived teammate's communication quality is above 0.9 for all the six times when this measurement was taken in this study, with an average of 0.935. Thus, we consider this scale reliable to use.

Teammate (Team Composition)	Round	Cronbach's Alpha Value
Alpha (HAA)	1	0.91
Alpha (HAA)	2	0.94
Bravo (HAA)	1	0.92
Bravo (HAA)	2	0.93
Charlie (HHA)	1	0.92
Charlie (HHA)	2	0.95
Human (HHA)	1	0.96
Human (HHA)	2	0.95
Average		0.935

Table 6. The Cronbach's Alpha Values of the Perceived Teammate Communication Quality

Perceived Teammate's Performance. Perceived teammate's performance measurement was adapted from an existing scale [23]. Nine five-point Likert scale items were used with the responses ranging from "Strongly Disagree" to "Strongly Agree". Example questions include: *The AI teammate Charlie I worked with did a fair share of the team's work*, *The AI teammate Charlie I worked with made a meaningful contribution to the team*, and *The AI teammate Charlie I worked with was actively involved in solving problems the team faced*.

Trust in the Teammate. Participants' trust in their teammates was measured through an existing validated Likert scale using six questions. This measurement was developed based on principles of trust identified by a previous study [69] and was rephrased in a way that participants could use a five-point Likert scale ranging from "Strongly Disagree" to "Strongly Agree" to answer. This measurement was taken twice after each round of the task, once for each teammate. Specifically, in the HHA team composition, the measurement was taken as (1) trust in the human teammate and (2) trust in the AI teammate. In the HAA team composition, the measurement was taken as (1) trust in AI teammate Alpha and (2) trust in AI teammate Bravo.

3.6.3 Team Measurements. Task load, team viability, and team performance were measured at a team level. Both task load and team performance were measured for each round of the task. Team viability was only measured once for each team composition.

Task Load. Task load was measured using a scale adapted from the NASA Task Load Index [49], which has been applied in human-AI teaming studies in previous work [80, 107]. Task Load Index is used to measure the subjective workload of tasks [49]. The adapted measurement removed the item related to "physical demand" due to the task characteristics of this study, and kept five items which were rated using a five-point scale ranging from "Very low" to "Very high". Based on Cronbach alpha's results, the fourth item was removed to ensure the reliability of this measurement.

Team Viability. Team viability is an important factor in predicting a team's sustainable growth and long-term success [9]. The scale used in this study was developed by Cooperstein in 2017 based on previous literature and measurements on team viability [22], which has been used in multiple studies since then [15, 120]. This measurement includes 14 questions that used a five-point Likert scale from "Strongly Disagree" to "Strongly Agree". Given that this measurement aims at predicting a team's long-term success, it was only taken once for each team composition. After participants finished two rounds of tasks with the same teammates (human&AI vs. AI&AI), they completed this measurement.

Team Performance. Team performance was measured in Rocket League as an objective evaluation of how the team performed in the tasks. The final score of each player was displayed on the screen after each round of the game. Specifically, Rocket League has a well-established scoring system,

which identifies specific actions and corresponding points to calculate the performance of each individual (e.g., scoring a goal, assisting/passing the ball to a teammate, blocking a shot). In this study, team performance was measured using the sum of each player's individual score (including both human and AI players). This measurement was measured at a team level (i.e., each team has the same team performance score each round). Participants who were on the same HHA team had the same team performance score.

3.7 Data Analysis

Two types of data were collected in this study, including task-related data (e.g., experiment task data and post-survey measurements) and the follow-up group interview data. To analyze the task-related data, we conducted linear mixed-effects models to explore the impact of AI's communication modalities (i.e., verbal and visual communication) on human perceptions (e.g., perceived communication quality, perceived teammate performance, trust) and objective team performance in various team compositions of triads (i.e., HHA vs. HAA). Moreover, some personal characteristics are included in the analysis, including participants' gender, previous Rocket League experience (i.e., experienced vs. inexperienced), and existing attitudes toward AI. Due to the nature of the two various team compositions, the individual-level measurement (i.e., trust, perceived communication quality) involved measurements toward the other human teammate in the HHA condition. Thus, we created a variable *Identity* as the identity of the teammate (i.e., human vs. AI).

The follow-up group interviews were analyzed using an inductive thematic analysis method [12] to achieve an understanding of communication patterns within various team compositions and participants' experiences collaborating with the AI teammate. The interview data analysis includes the following steps: (1) the first author read each interview transcript closely and highlighted words, phrases, and sentences that were related to the research question; (2) the first author then developed a preliminary set of codes and themes through an iterative process; (3) two of the authors discussed the themes and refine the themes; (4) the first author went through the transcripts again to extract quotes based on refined themes; (5) three of the authors further discussed the themes and finalized the themes to ensure a complete understanding of how humans perceive AI's verbal vs. visual communication in HATs similarly and differently.

4 Results

In total, 100 participants' valid data was collected, leading to 800 data points for the individual-level (teammate) measurement, and 400 data points for the team-level measurement. Additionally, 50 group interviews were conducted and recorded for the 100 participants. In this section, we present how our results address the research questions we proposed at the beginning of this paper.

4.1 RQ1: The Impact of AI's Communication Modalities on Team Coordination and Team Outcomes across Different Team Compositions

To address RQ1, we analyzed how AI teammates' communication modalities impact human-AI coordination in tasks using linear mixed-effects models. Specifically, the predictors include the manipulations of this study (i.e., AI's communication modalities and team compositions), personal characteristics (i.e., participants' gender, previous experience with Rocket League, existing negative attitudes toward AI), and two task-related attributes (i.e., task round and the identity of the teammate). The outcome variables include the post-measurements and task data (see Table 7). We will present the results of the impact of AI's communication modalities on team coordination at the teammate level and team levels.

Predictors	Manipulations	AI's Communication Modalities (Verbal vs. Visual) HAT's Team Composition (HHA vs. HAA)
	Demographic Information	Gender Existing Negative Attitudes Toward AI (NARS) Rocket League Experience (Inexperienced vs. Experienced)
	Task-Related	Identity of the Teammate (Only for Individual-level Measurements) Task Round (Round 1 vs. Round 2)
Outcome Variables	Teammate level	Perceived Teammate Communication Quality ⁽¹⁾ Perceived Teammate Performance ⁽¹⁾ Trust in the Teammate ⁽¹⁾
	Team level	Task Load ⁽²⁾ Team Performance ⁽²⁾ Team Viability ⁽³⁾

Table 7. Linear Mixed-effects Model Predictors and Outcome Variables

- (1) The variable was measured twice after each round of the task by each participant;
(2) The variable was measured once after each round of the task by each participant;
(3) The variable was only measured once for each team composition (i.e., after every two rounds of tasks).

4.1.1 The Impact of AI's Communication Modalities on Teammate-Level Measurements. In this subsection, we analyze the impact of AI's communication modalities on three teammate-level measurements, the perceived communication quality of teammates, including the perceived teammate performance of teammates, and the trust in teammates.

Perceived Communication Quality. The linear mixed-effects model of perceived teammate communication quality has a random intercept set for each participant. The model indicates that AI's communication modality has a significant impact on perceived communication quality ($b = -0.77$, $\chi(1)^2 = 20.70$, $p < 0.001$). Teammates in the verbal communication condition ($M = 4.03$, $SD = 0.95$) perceive higher communication quality than teammates in the visual communication condition ($M = 3.31$, $SD = 1.30$, see Figure 5).

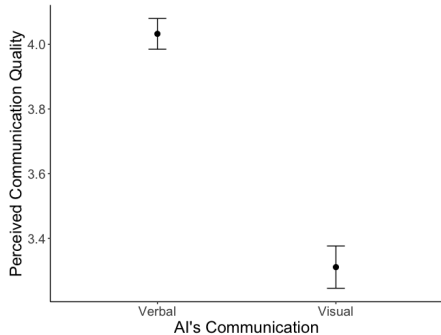


Fig. 5. Perceived Communication Quality by AI's Communication Modalities

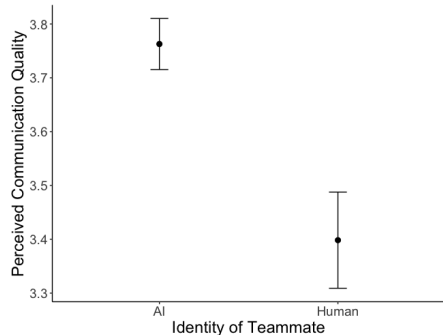


Fig. 6. Perceived Communication Quality of AI Teammate vs. Human Teammate

The model also shows that as tasks proceed, the perceived communication quality of teammates increases significantly ($b = 0.17$, $\chi(1)^2 = 6.50$, $p < 0.05$). People perceive their teammates as having

better communication quality in the second round ($M = 3.75$, $SD = 1.18$) compared to the first round ($M = 3.59$, $SD = 1.21$), regardless of HAT's team composition. The teammate's identity also significantly impacts the perceived communication quality ($b = 0.36$, $\chi(1)^2 = 24.29$, $p < 0.001$). Human teammates ($M = 3.40$, $SD = 1.26$) are perceived to have worse communication than AI teammates ($M = 3.76$, $SD = 1.16$, see Figure 6). This could be due to the difficulty of multitasking for human participants. Since humans have to use controllers to communicate (i.e., choose and send messages from the message list, see Table 5) and do the tasks simultaneously, they sometimes could not keep up with on-time communication, leading to worse perceived communication quality.

Participants' personal characteristics also have several impacts on the perceived communication quality. First, the linear mixed-effects model indicates that the impact of NARS on perceived communication quality depends on the participant's skill in Rocket League ($b = 0.86$, $\chi(1)^2 = 4.09$, $p < 0.05$). Specifically, for participants without prior Rocket League experience, the more negative their existing attitudes toward AI are (i.e., a higher NARS score), the lower perceived teammate communication quality they have. However, for participants who have previous experience with Rocket League, their existing negative attitudes toward AI (NARS) are positively related to the perceived communication quality (see Figure 7). In other words, the more negative they felt about the AI before the task, the higher communication quality they considered their AI teammates to have after the task. One explanation of this interesting finding is that experienced participants probably have more knowledge about game bots in general, and may even have previous experience teaming up with AI bots in Rocket League. However, the current bots in online multiplayer games are usually just a replacement for humans that are not good at communicating or collaborating with humans [128], which leads to low expectations of the AI's communication. Such low expectations may result in high satisfaction [92] when the AI performs better than expected.

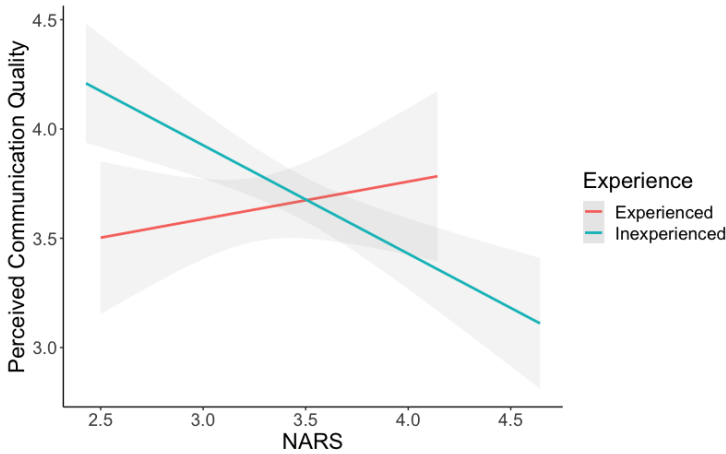


Fig. 7. Perceived Communication Quality of Experienced vs. Inexperienced Participants

Perceived Teammate Performance. A linear mixed-effects model with a random intercept set for each participant was developed. From the model, AI's communication modalities significantly impact perceived teammate performance ($b = -0.31$, $\chi(1)^2 = 9.44$, $p < 0.01$) of both human teammates and AI teammates. Specifically, participants who collaborated with AI using verbal communication perceive their teammates' performance as higher than participants who collaborated with AI using visual communication (see Figure 8).

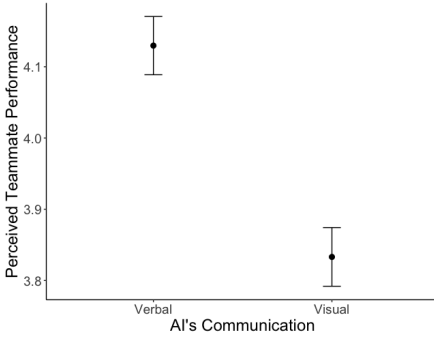


Fig. 8. Perceived Teammate Performance by AI's Communication Modalities

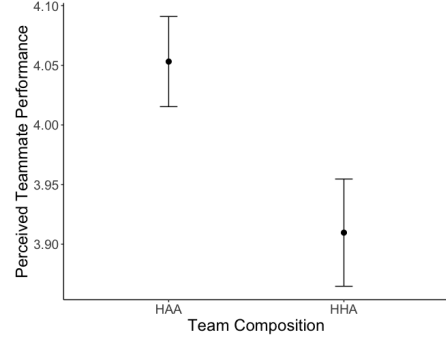


Fig. 9. Perceived Teammate Performance by Team Composition

Team composition also has a significant main effect on perceived teammate performance ($b = -0.23$, $\chi(1)^2 = 8.06$, $p < 0.01$). Participants in HAA teams perceive their teammates to have higher individual performance than in HHA teams (see Figure 9). This can be explained by the fact that AI's skills at the task are better than most participants. Thus, people who collaborated with two AI teammates perceived their teammates to have better performance. Moreover, the task round has a positive impact on perceived team performance ($b = 0.11$, $\chi(1)^2 = 5.94$, $p < 0.05$). Participants usually perceive their teammates to perform better in the second round ($M = 4.04$, $SD = 0.82$) than in the first round ($M = 3.93$, $SD = 0.84$, see Figure 10).

Participants' gender also has a significant interaction effect with the identity of the teammate on perceived teammate performance ($b = -0.27$, $\chi(1)^2 = 5.18$, $p < 0.05$, see Figure 11). While women participants perceive their human teammates to have better individual performance ($M = 3.63$, $SD = 0.92$) than men participants ($M = 3.32$, $SD = 1.17$), their perceptions of the AI teammate's performance are similar (women: $M = 4.14$, $SD = 0.71$; men: $M = 4.10$, $SD = 0.71$). The difference between women and men in perceived human teammate performance could be that women tend to be less confident than men about their competence [88], especially in team tasks [51], and they tend to undervalue their own contributions in human teamwork [50], which in turn may lead to overvaluing their human teammate's contribution.

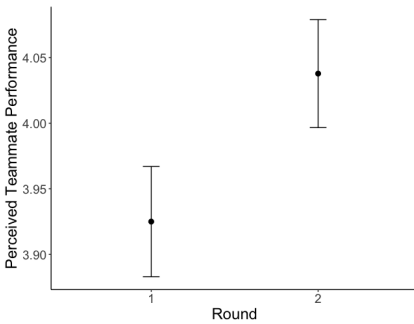


Fig. 10. Perceived Teammate Performance by Task Rounds

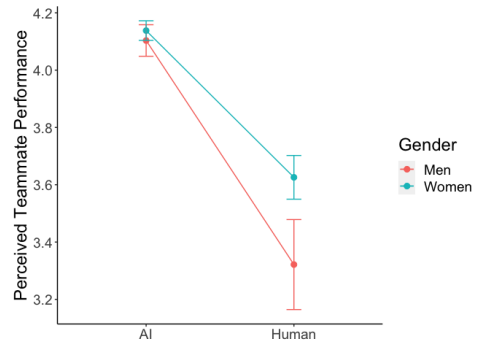


Fig. 11. Perceived Performance of Human Teammate vs. AI Teammate by Participants' Gender

Trust in the Teammate. We examined how much participants trusted each teammate (both human teammates and AI teammates) in two different team compositions and whether there was a difference between human teammates and AI teammates. The linear mixed-effects model's random intercept was set for each participant.

No significant main impact of AI's communication modalities on trust was observed from the model. However, AI's verbal communication trended towards receiving a higher trust than AI's visual communication (see Figure 12), which is similar to the trends of perceived communication quality and perceived teammate performance. The impact of HATs' team composition on trust in the teammate is dependent on the participants' NARS (Figure 13). People who have more negative attitudes toward AI trust their teammates less in HAA teams than in HHA teams, whereas people who have more positive attitudes toward AI trust their teammates more in the HAA team composition ($b = -0.17$, $\chi(1)^2 = 5.25$, $p < 0.05$). When people have negative perceptions of AI, teaming up with two AI teammates may generate a more negative collaboration experience, which then leads to lower trust in the teammate. However, people who have positive attitudes toward AI probably trust the two AI teammates (i.e., HAA) more than one AI teammate and one human teammate (i.e., HHA).

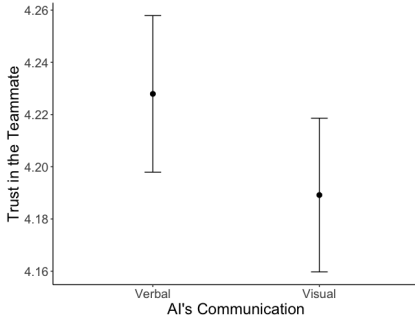


Fig. 12. Trust in the Teammate by AI's Communication Modalities [Non-significant Impact]

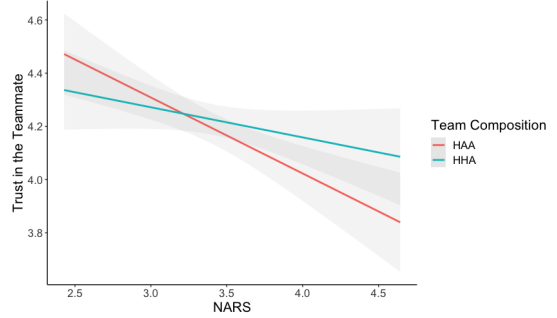


Fig. 13. Interaction Effects Between Team Compositions and NARS on Trust in the Teammate

The identity of the teammate also has a significant main effect on humans' trust in their teammate ($b = 0.24$, $\chi(1)^2 = 29.62$, $p < 0.001$). Participants trust human teammates ($M = 4.11$, $SD = 0.0.61$) significantly less than AI teammates ($M = 4.24$, $SD = 0.59$, see Figure 14). Since AI has advanced skills in team tasks, and most participants (68%) are inexperienced in Rocket League, humans are very likely to consider AI teammates' skills as higher than their human teammates' and trust the AI more. In addition, the impact of task rounds on trust in the teammate varies by participants' previous experience in Rocket League (see Figure 15). In the first round of tasks, inexperienced participants trust the teammate (whether the teammate was a human or AI) more than experienced individuals, but the difference is not significant in the second round ($b = 0.15$, $\chi(1)^2 = 5.35$, $p < 0.05$).

4.1.2 The Impact of AI's Communication Modalities on Team-Level Measurements. We analyze and present the impact of AI's communication modalities on three team-level measurements in this subsection, including task load, team viability, and team performance.

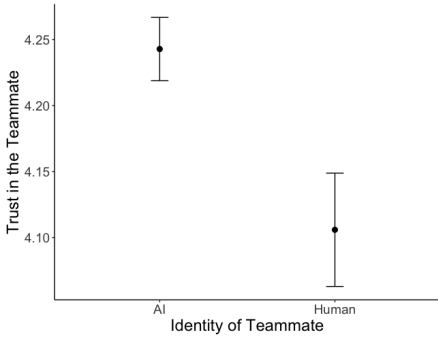


Fig. 14. Trust in the Human vs. AI Teammate

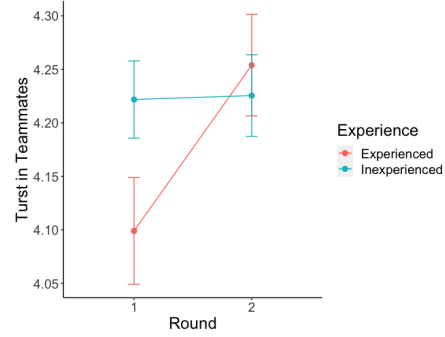


Fig. 15. Trust in the Teammate of Experienced/Inexperienced Participants by Task Round

Task Load. A linear mixed-effects model was built with task load as the dependent variable, and a random intercept set for each participant¹. From the linear mixed-effects model, AI's communication has a significant impact on perceived workload ($b = -0.30$, $\chi(1)^2 = 4.78$, $p < 0.05$). People who collaborated with AI using visual communication perceive the task as having a lower workload than people who collaborated with AI using verbal communication (see Figure 16). As expected, the task round has a significant effect on the perceived workload ($b = -0.08$, $\chi(1)^2 = 4.85$, $p < 0.05$). Participants perceive the workload to be lower in the second round than in the first round.

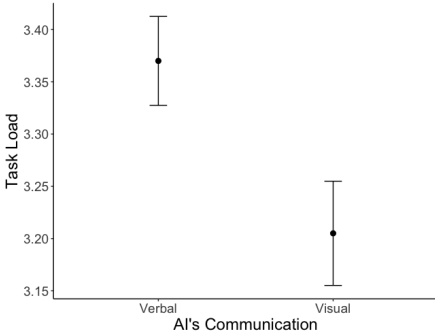


Fig. 16. Task Load by AI's Communication Modalities

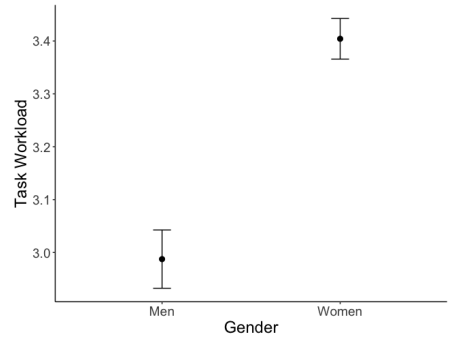


Fig. 17. Task Load By Participants' Gender

Personal characteristics also have two significant impacts. First, the gender of each individual significantly impacts their subjective workload ($b = 0.76$, $\chi(1)^2 = 18.28$, $p < 0.001$). Women tend to perceive the task to have a higher workload than men (see Figure 17). One possible explanation is that men are more familiar with competitive games [47]. Previous work also indicates that men prefer competitive games over cooperative games, whereas women prefer the opposite [87]. Such difference preferences may lead to men perceiving task load to be lower in competitive gaming environments than women. In addition, men usually have higher positive emotional responses

¹There were several data points that had high leverage (i.e., Cook's distance larger than 1). However, those points did not impact the linear mixed-effects model.

in competitive games [59], which accelerate the process of overcoming stresses [34]. Second, participants' NARS also have a significant impact on subjective workload ($b = 0.36$, $\chi(1)^2 = 9.59$, $p < 0.01$). When an individual has a more negative attitude toward AI, they perceive the task to have a higher workload.

Team Viability. The linear mixed-effects model of team viability has a random intercept set for each participant. No significant main or interaction impact of AI's communication modalities is observed on team viability. However, team composition has a significant interaction effect with people's NARS ($b = -0.58$, $\chi(1)^2 = 8.90$, $p < 0.01$) on team viability. While people's NARS is negatively associated with team viability, this negative association is stronger in HAA teams than in HHA teams (see Figure 18). People who have a more negative attitude toward AI (i.e., higher NARS score) perceive their HHA teams to have higher team viability than HAA teams. Having another human teammate on the team mitigates humans feeling disconnected from other teammates and encourages humans to be more collaborative in HHA teams [82, 105]. When humans have a stronger negative attitude toward AI, their disconnected feeling from their AI teammates would be stronger and may lead to a lower desire to collaborate with AI teammates. Since team viability indicates the long-term success of a team, it is likely to be low when humans have negative attitudes toward their AI teammates. Moreover, the impact of team composition on team viability also depends on people's previous experience with Rocket League ($b = -0.33$, $\chi(1)^2 = 5.58$, $p < 0.05$). In particular, experienced participants perceive the team ($M = 3.77$, $SD = 0.46$) as having higher team viability than inexperienced participants ($M = 3.52$, $SD = 0.74$) in HHA teams, whereas no significant differences in HAA teams (see Figure 19). Each individual member's experience and skill are considered in evaluating the growth needed to achieve future team success. Experienced participants probably are more confident in the future success of the team due to their previous experience with Rocket League.

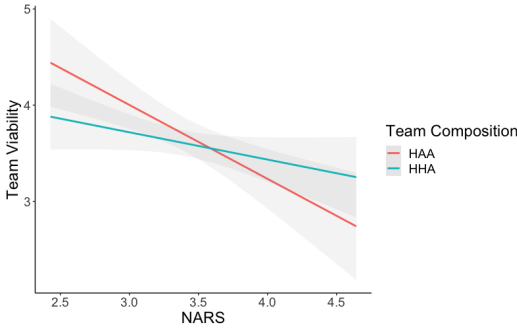


Fig. 18. An Interaction Effect of Team Composition and NARS on Team Viability

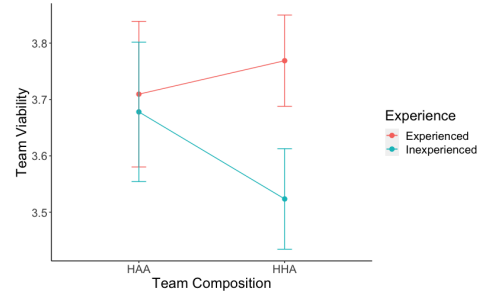


Fig. 19. An Interaction Effect of Team Composition and RL Experience on Team Viability

Team Performance. As the raw team performance score was not normally distributed, it was transformed using a $\log()$ function to ensure that the transformed team performance was normally distributed. We then built a linear mixed-effects model with the transformed team performance score and a random intercept set for each participant. The linear mixed-effects model shows a significant interaction effect between AI's communication modalities and team composition on transformed team performance scores ($b = -0.06$, $\chi(1)^2 = 4.38$, $p < 0.05$). While there is no significant difference in team performance for AI's verbal and visual communication in HAA teams, AI's visual communication benefits team performance more than verbal communication in HHA

teams (see Figure 20). As expected, team performance has a significant increase as tasks proceed ($b = 0.04$, $\chi(1)^2 = 11.91$, $p < 0.001$). Participants perform better as a team in the second round ($M = 7.62$, $SD = 0.14$) than in the first round ($M = 7.58$, $SD = 0.11$).

Moreover, there is a significant interaction effect between team composition and participants' gender on team performance ($b = 0.08$, $\chi(1)^2 = 7.72$, $p < 0.01$). In particular, men have better team performance in HHA teams ($M = 7.65$, $SD = 0.14$) than women ($M = 7.58$, $SD = 0.12$), whereas the differences of gender are not significant in HAA teams (men: $M = 7.61$, $SD = 0.14$; women: $M = 7.60$, $SD = 0.12$), as shown in Figure 21. Since humans could get less collaborative in HAA teams than HHA teams [82], men could be competitive to get better individual performance scores than AI, leading to internal competition and lower team performance in HAA teams than in HHA teams. Additionally, men's average individual performance is higher than women's (women: $M = 3.85$, $SD = 0.90$; men: $M = 5.31$, $SD = 0.75$), which is likely to contribute to higher team performance.

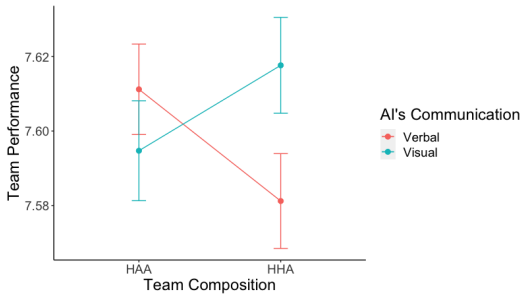


Fig. 20. Team Performance by AI's Communication Modalities in Two Team Compositions

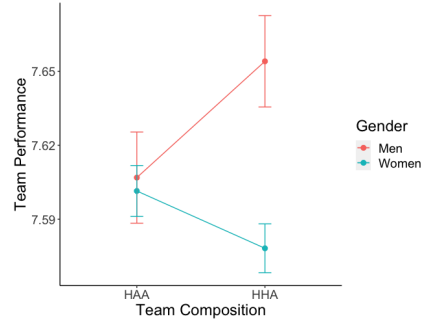


Fig. 21. Team Performance Score by Participants' Gender in Two Team Compositions

4.1.3 Summary. The experimental analysis generates multiple significant findings that are summarized and presented in Table 8. Specifically, there are three main findings related to AI's communication modalities: (1) there is no significant impact of AI's communication modalities on human trust or team viability; (2) AI's visual communication decreases perceived workload but was perceived less positively than AI's verbal communication (e.g., lower communication quality and lower perceived teammate performance); and (3) AI's visual communication contributes to higher team performance than AI's verbal communication in HHA teams.

Outcome Variables	Significant Predictors ⁽¹⁾
Perceived Communication Quality	CommModa ⁽²⁾ *** + Task Round * + Identity *** + Experience:NARS *
Perceived Teammate Performance	CommModa ** + TeamComp ⁽²⁾ ** + Task Round * + Identity:Gender *
Trust in the Teammate	TeamComp:NARS * + Identity *** + Experience:Task Round *
Task Load	CommModa * + Task Round ** + NARS ** + Gender ***
Team Viability	TeamComp:NARS * + TeamComp:Experience * + NARS **
Team Performance	CommModa:TeamComp * + TeamComp:Gender ** + Task Round ***

Table 8. The Significant Predictors of Each Outcome Variable

Notes:

(1) When there is both an interaction effect and a main effect of a predictor on an outcome variable, only the significant interaction effects are included in the table (* $p < .05$, ** $p < .01$, *** $p < .001$).

(2) CommModa refers to AI's communication modalities; TeamComp refers to HATs' team compositions.

4.2 RQ2: How Humans Perceive AI's Verbal vs. Visual Communication Similarly or Differently

Overall, the quantitative results indicate that AI's verbal communication is perceived more positively than AI's visual communication, but AI's communication modalities do not appear to impact human trust. In this section, we further explore the similarities and differences in human perceptions of AI's verbal vs. visual communication in HATs.

4.2.1 Both AI's visual and verbal communication could contribute to trust increase under certain criteria. Even though no significant impact of AI's communication modalities on trust was found in the quantitative analysis, both AI's verbal and visual communication could contribute to securing trust when certain criteria are met. We identified two criteria that AI is expected to meet.

First, AI need to adjust their actions based on humans' communication. Humans expect their communication with AI to change AI's decisions and actions. When such communication leads to little or no impact, communication from AI, in turn, plays little role in impacting human trust in HATs. Participants from both communication conditions emphasized the importance of such communication-based change in AI's behaviors:

I don't think it (AI's communication) contributed to my trust in them that much, because they were just doing whatever they wanted to do. So like he said, my trust in that communication declined over the four games. (T40, PB, Woman, Verbal)

When I told them at first what I was going to do, they let me do it, and then they would come and help. They trusted me as well. But in the last two rounds, even when I communicated, they just kind of disregarded it. So it just made me think, oh, they're just kind of doing their own thing. I still trusted them because I knew that they could score but it was just a little different. (T23, PA, Woman, Visual)

According to these quotes, when humans' communication has little impact on AI's actions, it is likely to harm their trust in AI's communication regardless of AI's communication modalities. Rather, when AI seems to take humans' communication, it is perceived as *the AI teammate trusts the human*. Such trust regarding AI's communication should be differentiated from another type of trust that was developed on AI's capability in accomplishing tasks. Even when AI's communication does not receive human trust, AI's advanced task skill still gains human trust.

Second, AI's communication should present their cognitive thinking and decision-making process in communicating with humans. For instance, PA in T31 shares that:

It would have been a little bit different if it was not just pre-made messages, if they were actually communicating, like typing the message. But because it was a robot, their communication automatically change whatever they were doing. Like me, I would have to communicate what I was doing. It's a lot harder doing that. But for them, it almost seems like it's automatic. It would change every time they're doing something different. (T27, Woman, PA, Verbal)

For PA in T31, AI's messages are automatic, which does not present their efforts in communication, as humans do. Rather, AI typing messages is considered to take more effort than AI sending pre-developed messages, even though AI is obviously coded to send certain messages due to their machine and automatic feature. Such automatic feature of AI's communication in this case weakens the impact of AI's communication on trust. Similar to AI's verbal communication, AI's visual communication is perceived as automatic and cold coding:

So you couldn't really trust it, since it was just based on coding and facts. I wouldn't trust just a trajectory. That's just the fact that they're going in different directions. It's not based on their own decision. (T29, PA, Man, Visual)

For PA in T29, AI's visual communication does not reflect their decision-making process. Such visual representation is incapable of presenting the information as text messages do, which indirectly shows the decision process. PA emphasizes the importance of AI sharing their cognitive process, which is crucial in impacting human trust.

In summary, both AI's verbal and visual communication has the potential to benefit human trust, but only (1) when AI adapts to human communication, and (2) when their communication shows that they endeavor to make a decision which then is communicated using either verbal or visual communication.

4.2.2 AI's visual communication alone lacks of features that "real communication" has. While humans use natural language to communicate, AI, with their special machine feature, is expected to use other communication modalities [128]. However, our interview data indicate that AI's visual communication is perceived as worse than AI's verbal communication for two reasons. On the one hand, **AI's visual communication, as the only communication modality, is not considered as the type of communication humans define** since it lacks verbal communication's interactive feature. For instance, some participants shared that:

It was kind of hard to adapt, since there was no communication on their end. Obviously, I tried communicating, but like they didn't communicate back. (T32, PA, Woman, Visual)

There's definitely a lack of communication, but they're really good at the game. (T25, PB, Woman, Visual)

Both quotes indicate the insufficiency of AI's visual communication. In particular, PA in T32 emphasized the importance of the "interactive" feature of communication. Such interactive feature is natural to humans in communication:

The communication is hard because all you're really going off of is their lines. And you can send stuff, but you can't get a response. The only communication you have is like the directory lines. That was the hardest aspect of the first few rounds. (T33, PB, Woman, Visual)

Without text-based communication, AI's visual communication is uni-directional communication. PB in T33 pointed out that bi-directional communication, through which team members can exchange information and coordinate accordingly, is crucial for them to adapt to the game, especially at the beginning of the task. Thus, it is difficult to utilize such visual communication to coordinate effectively:

Sometimes I like some of the times they were pretty accurate and when helped me but other times there's confusing. They would take too long and I would be there too soon, or something and then it just won't work out. (T31, PB, Woman, Visual)

On the other hand, **it is challenging to coordinate with AI teammates that use visual communication.** First, there is a gap between the presentation of visual communication (i.e., AI's trajectory) and the real meaning behind it, which takes extra cognitive effort for humans to connect the two:

I don't think they really communicated that well. I saw what they were doing, but I didn't really know what they were doing. (T30, PB, Woman, Visual)

It made it really hard to say what they were going to be doing and then they would just not say anything, and then take over. And they bump into me. (T49, PA, Woman, Visual)

According to these two quotes, AI's visual communication was not well understood by humans and thus could not be utilized in their coordination. Specifically for PA in T49, AI's visual communication was not considered as communication, as how they described "they would not say

anything”. This perception of AI’s visual communication brings new insights into how AI should be designed using various types of communication modalities. Second, compared to AI’s verbal communication, AI’s visual communication has limitations on the visibility of shared information:

Since there are no text communications, if I was looking at the other team’s goal, I couldn’t see the lines, and I didn’t know where my two teammates were. So I couldn’t really rely on them. I don’t think the green lines were the best method of communication. It was good in some scenarios when I was able to see them. But for the most part, I couldn’t really rely on them. (T34, PB, Man, Visual)

Even though AI’s visual communication provides more spatial information, it indeed has worse visibility than AI’s verbal communication in certain scenarios, where AI’s communication trajectory is not within humans’ eyesight, making it more difficult to coordinate. Another participant shared a similar opinion on the visibility of AI’s visual communication:

You know what their next step is, but you don’t know how fast they’re gonna get to the ball, or if you’re gonna get there first. Obviously, it [communication] makes it easier to work together and collaborate, but since there’s no communication (from AI) either, that’s also what makes it harder to (collaborate). We ran into each other many times because of that. However, the trajectory definitely helps rather than not having. (T32, PA, Woman, Visual)

As PA in T32 pointed out, the information provided through the AI’s visual communication was not enough for human players to react effectively. In such a fast-paced environment, this lack of information is a road blocker in their coordination. When the visual information is not visible, it is difficult to maintain an awareness of AI’s status, which is essential to achieving high team performance.

In summary, AI’s visual communication was perceived more negatively compared to AI’s verbal communication. First, AI’s visual communication lacks the interactive feature that natural-language-based communication has, which is important for humans to incorporate into team coordination. Second, participants found it difficult to use AI’s visual communication to support their coordination with AI teammates due to its lack of visibility at times and lack of naturalness.

5 Discussion

This study explores how AI’s use of verbal and visual communication impacts team coordination and team outcomes across two team compositions. Using an experiment and a follow-up group interview with 100 participants, our findings highlight: (1) AI’s visual communication decreases subjective workload but is perceived as lower communication quality and lower individual performance, compared to verbal communication (RQ1); (2) no significant impact was found from AI’s communication modalities on trust in the teammate, or team viability across various team compositions (RQ1); (3) AI’s visual communication yields significantly better team performance than AI’s verbal communication in HHA teams (RQ1); (4) AI’s verbal/visual communication could contribute to trust under certain conditions, including (a) *AI need to adjust their actions based on humans’ communication* and (b) *AI should present their cognitive thinking and decision-making process in their communication* (RQ2); and (5) unlike verbal communication, AI’s visual communication lacks features that make the message easy to understand, such as naturalness and interactivity (RQ2). These findings highlight the importance of a nuanced understanding of the trade-offs of AI’s visual and verbal communication. In the remainder of this section, we provide a thoughtful discussion of these trade-offs and how they can be leveraged to design AI communication capabilities for HATs. In doing so, we also acknowledge the limitations of our work and identify potential avenues for future research to explore.

5.1 Trade-offs of AI's Visual and Verbal Communication in Team Coordination

Our results present multiple significant impacts of AI's communication modalities on team coordination in HATs. In this section, we further discuss the trade-off that AI's visual and verbal communication has, as well as design recommendations on applying a combination of AI's visual and verbal communication to maximize the efficiency of AI's communication in supporting human-AI coordination in teams.

5.1.1 New Perspectives of AI's Visual and Verbal Communication in HATs. Our experiment results and group interview findings highlight three new insights into AI's use of visual and verbal communication in HATs: (1) AI's use of visual communication only contributes to team performance in HHA teams; (2) AI's visual communication is perceived worse compared to AI's verbal communication in HATs; and (3) both AI's verbal and visual communication could contribute to human trust under certain conditions.

First, while AI using visual communication decreases subjective workload in both HHA and HAA teams, it only benefits team performance in HHA teams (see Figure 20). We suspect that this is likely due to humans' inability to effectively leverage the emancipated cognitive capacity to collaborate with AI teammates. Previous work indicates that humans are more collaborative in HHA teams than in HAA teams due to AI outnumbering humans [82], partly because it is difficult to build shared understanding with and collaborate with AI teammates [104]. As a result, humans often feel more "disconnected" working with only AI teammates and no other humans, which can be mitigated by adding another human teammate [105]. It is therefore possible that humans may have had more negative collaboration experiences in HAA teams than in HHA teams, which then may have offset the benefits of AI's visual communication on workload. More importantly, humans who teamed up with one AI teammate may make better use of AI's visual communication to coordinate implicitly. In contrast, two AI teammates' visual communication (i.e., two trajectories) may cause more confusion through the dynamic teaming environments in HAA teams (e.g., "*It became more confusing having two green lines coming from different directions.*").

Second, even though AI's visual communication could contribute to higher team performance in HHA teams, humans seem to have more negative perceptions of AI's visual communication and are unaware of visual communication's benefits. In particular, humans perceive AI with visual communication as having lower communication quality (see Figure 5) and lower individual performance (see Figure 8) than AI with verbal communication. As detailed in the Method section, all the AI teammates in this experiment apply the same play strategy (i.e., using the same code-based algorithm). Thus, all the AI teammates have the same objective individual performance in performing the team tasks. One possible reason for the negative perceptions of AI's visual communication is that their visual communication does not align with humans' traditional definition of communication. Previous work defines communication as "the discriminatory response of an organism to a stimulus" [110]. In other words, when the teaming environment has certain stimuli (e.g., a need of sharing information or requesting information) to a teammate (i.e., human or AI), they would provide a discriminatory response. This then requires each individual's communication to be bi-directional, which is important in developing shared mental models, human trust in the AI, and team-level trust in HATs [30]. However, AI's visual communication is uni-directional and lacks the capability of providing immediate responses. In our group interviews, individuals indicated their preference for AI using natural-language-based communication with flexibility, because AI's visual communication was considered as "codes" and "facts", which is not *real* communication to humans. More importantly, our study shows that AI's cognitive thinking process is expected to be included in their communication, which is more challenging for AI's visual communication to present than AI's verbal communication. Last but not least, AI's visual communication in a

3D space has a visibility limit. When it is not received appropriately (e.g., not within humans' eyesight on the screen) during their collaboration, it could result in a lack of communication that hinders coordination. Especially in stressful situations [123], breakdowns in coordination due to lack of communication often lead to poor team performance. All these limitations of AI's visual communication may lead to a negative perception.

Last, our work shows that AI using verbal or visual communication could contribute to trust under certain conditions. When humans believe their communication can change AI's behaviors and course of action, AI's communication is more likely to be taken seriously by humans. This emphasizes the strong connection between AI's communication and their expected behaviors, as well as the importance of designing adaptive AI [28]. Additionally, AI is expected to present their cognitive thinking process or decision-making process in their communication, regardless of the communication modality AI uses. This aligns with the core concept of transparency and explainability in previous work: AI needs to generate effective explanations of their decision-making models [42]. Interestingly, our interview data show that even AI typing messages out could be used as AI showing their "cognitive thinking process". Even though AI is known for their machine nature, humans still expect AI to mimic humans' behaviors, for facilitating the kind of collaboration they are accustomed to with human teammates [128].

5.1.2 Design Recommendations. Driven by these new insights into the trade-offs between AI's use of visual and verbal communication, we suggest that purposefully combining AI's verbal and visual communication to complement one another will increase AI's communication effectiveness.

First, *AI's visual communication should be applied to share information that may be less effectively conveyed through text-based communication*, such as information about shared spatial environments, where visual cues communicate rich information to increase the efficiency of team communication. Communicating spatial information quickly and accurately is especially beneficial to contexts that involve time-sensitive tasks in 2D and 3D environments, such as reconnaissance and rescue tasks commonly adopted in practical HAT settings [19, 108]. These tasks require humans and/or AI to communicate various types of spatial information (e.g., locations and directions) and complete tasks effectively. Importantly, our work has shown AI's visual communication is perceived as requiring a lower workload, and contributes to better team performance than verbal communication in HHA teams. These strengths of AI's visual indicate the essential role of visual communication in shared spatial environments. Previous work on communication and coordination in *human-only teams* also suggests visual communication be an effective way to exchange information in a shared visual space [64]. In addition, spatial information communication is broadly used in virtual reality and mixed-reality environments [72, 97], in which shared spatial environments are created for users and AI agents. Using visual communication in shared spatial environments benefits HATs by increasing humans' shared awareness of each team member's progress [106] and implicitly coordinating with AI teammates. Previous work has pointed out that shared awareness plays a crucial role in both human-human teams and human-AI teams [40, 70], especially in developing shared mental models [44, 105]. With the advancements in technologies and the advantages of visual communication, the application of visual communication could increase human-AI communication effectiveness and facilitate the development of shared mental models in HATs, which contributes to high team performance [73]. Notably, our work emphasizes a new insight- AI's visual communication needs to be visible to humans at all times to ensure communication effectiveness in time-sensitive tasks. This holds particular significance in an era marked by rapid technological advancements that enable increasingly intricate and multidimensional virtual collaborative environments.

Second, *AI's verbal communication should be implemented to meet human communication requirements, especially in facilitating conversations with humans*. Our findings emphasize the importance of

AI being responsive to humans' messages and AI being able to build interactive conversations with humans. Verbal communication should be applied for humans' comfort in team communication. Specifically, AI teammates are anticipated to give feedback and show their cognitive processes using verbal communication. While visual communication is efficient in sharing certain information, the conversational feature of verbal communication plays a critical role in building trust and facilitating team processes. Previous work on AI's communication has shown that AI with proactive communication assists trust development and situational awareness [126]. Combining AI's verbal and visual communication with the above implementation recommendations will maximize AI's communication efficiency and also meet humans' communication needs, which helps build smooth and satisfying human-AI communication HATs.

5.2 Generalizability and Future Applications

While this study presents a deep mixed-methods narrative exploring communication in human-AI teams, considerations need to be made to explicitly highlight the generalizability of this work. In particular, the use of Rocket League as an experimental platform yields a research paradigm that is not fully representative of the broader teaming landscape. In turn, it is important to highlight the aspects of Rocket League that potentially influence the results of this work, and how alternative contexts may see similar or different results.

5.2.1 Rocket League Teaming Environment. While using Rocket League as a research platform may have influenced participants' preference for communication, identifying specific features Rocket League has as a teaming environment provides insights into the external application of this study's findings. In this subsection, we first summarize some features of Rocket League as a teaming environment and then discuss how these features may impact communication preferences. Following this, we discuss how the findings may be applied in other experiment platforms.

The features of Rocket League are identified through the lens of teamwork based on previous literature [114, 126], shown in Table 9. One important characteristic of the Rocket League teaming environment is that HATs were set up in a competitive gaming environment with rapidly changing team status. Participating in a virtual sports game in such a teaming environment greatly impacts team communication and human perceptions of said communication. In particular, Rocket League presents a competitive and continuous task where teammates are constantly moving, acting, and communicating. Both the competitive context and continuous nature may ultimately have led teams to favor communication with less overhead [71], as any communication that distracts one from the continuous task may feel antithetical to the competitive goal in front of them. As it relates to our results, this feature may have led to a stronger preference for verbal communication, as quick verbal pushes can quickly communicate team members' intent and actions.

Similarly, not all team compositions are the same, and while this study did leverage two different team compositions, team-specific considerations still exist. Notably, Rocket League presents a type of self-organizing team, where distinct roles are not pre-arranged but rather rotated in and out as the task proceeds. Within teamwork, self-organizing teams are not uncommon, but they are unique teaming constructs as most teammates in these teams usually share similar goals and information. In turn, over-redundancy can become problematic to these teams in dividing work and responsibilities [81], and teammates need to consistently ensure that they are working interdependently and not on the exact same workload. This premise can also be seen in Rocket League, as team conflict could occur if both teammates prioritize the exact same roles and actions within the field. As such, teammates may feel more comfortable working with a more explicit communication modality, such as verbal communication, to prevent this redundancy.

Table 9. Rocket League Teaming Environments

Dimension Type	Dimension	Details
Task	Task Type	A competitive task with opponents.
	Situational Stressors	High stress: Five-minute timed task with a timer and a goal scoreboard displayed on the screen
Team	Roles	Humans and AI teammates all share the same responsibilities in team tasks.
	Size/Composition	Two types of HAT: HHA and HAA . Each HAT has three team members.
Communication	Human Communication Modality	A text-based chat channel.
	AI Communication Method	Two types of AI modalities are provided using a between-subjects design: (1) AI using verbal communication (text-based chat channel); (2) AI using visual communication.

Lastly, the actual communication structure within Rocket League was also unique in that it either consisted of unidirectional verbal communication or unidirectional visual communication. In turn, the overhead related to communication was naturally lowered, as bi-directional communication was not required or normalized. Similar to task-related aspects, this simplification in communication could have lowered the general burden of both visual and verbal communication methods. However, when team tasks require humans to increase their verbal communication, it could become a burden that harms human-AI teamwork by distracting humans from their own responsibilities.

5.2.2 Future Application and Research in Other HAT Contexts. Outside of the Rocket League context, the results of this work may see consistency or difference based on contextual differences. Highlighting similar contexts, preferences for verbal communication would likely persist in environments where tasks are continuous, teams rely on self-organization or communication overhead is generally low. For example, as AI progresses in the field of software development, which often sees teams self-organize [81], verbal communication may be consistently preferred to ensure that the integration of AI teammates does not ultimately lead to residency in workload. Similarly, emergency response and medical contexts may also see a more explicit preference for AI teammates that use verbal communications, as teams in these contexts often try to minimize communication overhead while accurately communicating intent in a fast-paced and continuous work environment [85].

Teaming environments that deviate from these features may starkly differ. For example, multi-AI environments that leverage large numbers of AI technologies, such as drone swarms, may be overly burdened by verbal communication [99], leading to a preference for visual communication. Indeed, prior research has shown that some visual information can be beneficial to human-AI teams that perform resource management in spatial environments [105], and these teams may be inadvertently burdened by an overreliance on verbal communication. Alternatively, teams that have very explicit roles and see a clear division of responsibility may have a preference for visual communication as a primary means of communication, as teammates would likely want to avoid the pushing of irrelevant information and focus more on understanding an AI teammate’s status. However, for CSCW/HCI research to fully understand human preferences and perceptions of different communication modalities within human-AI teams, additional work will still be needed across different contexts.

5.3 Limitations and Future Work

As future work continues to explore human-AI teaming, the results and limitations of this work should serve to inspire future efforts. First, while this study endeavors to generate insights that can be applied in other contexts, this study examines AI's verbal and visual communication using one type of task within a specific context. Due to the specific task and context characteristics, the findings may be different in other tasks or contexts. The discussion over the future application of this study's findings provides an initial step for generalizing the results. Nevertheless, future work should compare their target context against this context and develop their research plans based on the comparison and investigate how AI's verbal and visual communication is perceived and how it impacts team coordination in other contexts. Second, all participants are undergraduate students from a local university. This young population may perceive and collaborate with AI agents in a different way from other populations, such as senior adults. Future research should examine and extend the findings in this study to a population with more diverse backgrounds. Last, this study uses a pre-developed list of messages for AI teammates. While this ensures the consistency of triggered events and AI's communication content between AI's verbal and visual communication conditions, future work may employ a flexible AI message list using advanced natural language processing models.

6 Conclusion

In this study, we examine the impact of AI's communication modalities (i.e. text-based verbal communication vs. visual communication) on human perceptions and team outcomes in HATs with two team compositions (i.e., HAA vs. HHA). Our findings uncover a trade-off between the advantages and disadvantages of AI's verbal and visual communication in human-AI teams: while AI's visual communication decreases subjective workload and contributes to team performance in HHA teams, it is perceived as lower communication quality due to its constraints, such as lack of interaction and limited visibility in 3D space. However, by balancing the advantages and disadvantages of each communication modality, AI's communication could be designed with both efficiency and human satisfaction. This work generates insights into the role of AI's communication modalities in supporting team coordination and generating team performance, which future research on human-AI communication can build upon to further understand the application of AI's communication in HATs.

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